

Submit your assignment solution on Canvas. You may discuss with others or seek help from your TA, but you should not directly copy from others. Otherwise, it will be considered as plagiarism.

(1) Bayes' theorem

A telegraphic communications system transmits dots and dashes. Assume that due to noise, an average of $2/5$ of the dots and $1/3$ of the dashes are changed. Suppose that the ratio between the transmitted dots and the transmitted dashes is $5 : 3$.

- (a) What is the probability that a received signal will be the same as the transmitted signal if ...
 - (i) ... the received signal is a dot?
 - (ii) ... the received signal is a dash?
- (b) Assuming independence between signals, if the message dot-dot was received, what is the probability distribution of the four possible messages that could have been sent?

(2) Characteristic function

A random variable X has probability density

$$f(x) = \frac{1}{2} e^{-|x|}$$

- (a) Find its characteristic function.
- (b) From the characteristic function, find its mean and variance.

(3) Autocorrelation

Let $X(t)$ be a stationary random function whose values are normally distributed with zero mean and variance σ^2 , then let

$$Z(t) = \frac{1}{2} \left[1 + \frac{X(t)X(t + \tau)}{|X(t)X(t + \tau)|} \right]$$

for some lag τ . Show that the time average of $Z(t)$ is

$$\bar{Z}(\tau) = \frac{1}{\pi} \arccos \left[-\rho(\tau) \right]$$

where $\rho(\tau)$ is the normalized correlation function of $X(t)$. [Hint: Let $x = X(t)$ and

$y = X(t + \tau)$ be two normal random variables and consider their joint distribution.]

(4) Radar system

A radar system uses radio waves to detect aircraft. After receiving a signal, the system needs to decide whether an aircraft is present or not. Let X be the received signal. We know that

- $X = W$ if no aircraft is present
- $X = 1 + W$ if an aircraft is present

where $W \sim N\left(0, \sigma^2 = \frac{1}{9}\right)$ is some inevitable noise. Thus, we can write $X = \theta + W$,

where $\theta = 0$ if there is no aircraft and $\theta = 1$ if there is an aircraft. Now we define the null hypothesis H_0 and the alternative hypothesis H_1 as

- H_0 : no aircraft is present
- H_1 : an aircraft is present

- (a) Write H_0 and H_1 in terms of possible values of θ .
- (b) Design a test to decide between H_0 and H_1 by setting its significance level as $\alpha = 0.05$.
- (c) Find the probability β of type II error for the test in (b).
- (d) If we observe $X = 0.6$, is there enough evidence to reject H_0 at significance level $\alpha = 0.01$?
- (e) If we would like the probability of missing a present aircraft to be less than 5%, what is the smallest significance level that we can achieve?
- (f) Design another test to decide between H_0 and H_1 by comparing the likelihood ratio of the two hypotheses with some preset threshold η .